



# **TERMS OF REFERENCE (TOR)**

**DZONGJUK**

**DZONGKHA STANDARD TESTING SYSTEM(DSTS)**

**DEPARTMENT OF CULTURE AND DZONGKHA DEVELOPMENT**

**MINISTRY OF HOME AFFAIRS**

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## **1. Introduction**

This Terms of Reference (ToR) outlines the requirements for a software development project. It serves as a document for prospective consultants, detailing the scope, objectives, deliverables, and expectations for the engagement. The purpose of this document is to invite proposals from qualified software development consultants to design, develop, and implement a software solution that meets the specified functional and non-functional requirements.

Consultants are expected to thoroughly review this ToR, along with the accompanying functional and non-functional documentation, to formulate comprehensive and competitive proposals. By submitting a proposal, the consultant acknowledges that it has conducted its own independent due diligence and has not relied on any representations, promises, or warranties by the Procuring Agency except as expressly set forth in this ToR.

## **2. Project Background and Objectives**

### **2.1. Project Background**

Bhutan's cultural identity, sovereignty, and social cohesion are deeply rooted in the Dzongkha language. However, there currently exists no unified national standard for assessing Dzongkha proficiency. Multiple institutions conduct independent examinations, resulting in inconsistent standards, duplicated efforts, and inefficient use of public resources.

To address this, the Department of Culture and Dzongkha Development introduced the DSTS, drawing upon global best practices such as IELTS and CEFR, and local assessment frameworks. The growing national importance of Dzongkha across governance, education, citizenship, media, judiciary, and public service demands a robust digital system capable of supporting standardized, scalable, and sustainable assessment operations.

### **2.2. Project Objectives**

The primary objectives of this software development project are to:

- 2.2.1. Objective 1: Provide a centralized digital mechanism to conduct DSTS Test and manage examination results, band scores, and certificates.
- 2.2.2. Objective 2: Ensure timely, accurate, and transparent publication of results.
- 2.2.3. Objective 3: Enable controlled access to band scores and result status for test takers.
- 2.2.4. Objective 4: Digitize the storage and access of issued certificates for long-term reference.
- 2.2.5. Objective 5: Improve administrative efficiency and reduce dependency on manual record-keeping.
- 2.2.6. Objective 6: Support re-evaluation outcomes and approved score revisions within the system.

### **3. Scope of Work and Deliverables**

#### **3.1. Scope of Work**

The scope of work for this project encompasses the full software development lifecycle, from detailed design to deployment and support.

- 3.1.1. Requirements Analysis: Detailed understanding and validation of functional and non-functional requirements. Detailed non-functional, and functional requirements (NFRs) are outlined in the accompanying requirements document.
- 3.1.2. System Design: Architectural design, database design, and user UI/UX design.
- 3.1.3. Development: Coding, unit testing, and integration of all solution components.
- 3.1.4. Quality Assurance: Comprehensive testing, including system testing, user acceptance testing (UAT) with scripts, and performance testing.
- 3.1.5. Deployment: Installation, configuration, and launch of the software.
- 3.1.6. Documentation: Creation of technical documentation, user manuals, and training materials.
- 3.1.7. Training: Training for end-users, system administrators and relevant stakeholders.
- 3.1.8. Post-Implementation Support: Warranty period support as per the SLA.

#### **3.2. Deliverables**

The successful consultant will be responsible for delivering, including:

- 3.2.1. Detailed Project Plan: A comprehensive plan outlining timelines, milestones, resource allocation, and risk management strategies.
- 3.2.2. Design Documents: Architecture Document, Database Schema, UI/UX and Mockups.
- 3.2.3. Source Code: Well-documented and clean non obfuscated source code for the entire software.
- 3.2.4. Test Plans and Reports: Documentation of testing procedures, results, and bug resolution.
- 3.2.5. Deployed Software: A fully functional and tested software application with warranty.
- 3.2.6. User Manuals and Training Materials: Comprehensive guides for end-users and administrators. Supply training materials, tutorials (video or documentation) and training schedules.
- 3.2.7. Technical Documentation: API documentation, UX/UI, deployment guides, and maintenance procedures.
- 3.2.8. Training Sessions: Conducted for identified stakeholders by Procuring Agency.

### **4. Reporting Structure**

The Consultant is responsible for maintaining transparent and regular communication with the Procuring Agency's Project Focal. This includes:

- 4.1. Progress Meetings: Participation in scheduled progress meetings. Mandatory submission of detailed status reports covering progress, completed tasks, planned activities, and any identified issues or blockers.
- 4.2. Risk Reporting: Immediate reporting of any critical risks or issues that could impact the project timeline or budget. The Consultant must establish and maintain a Risk Register, which continuously captures potential risks, assesses their impact, and details mitigation strategies.

## 5. Data Handling and Confidentiality

The Consultant must adhere to strict protocols regarding data handling and confidentiality, as the system will manage sensitive citizen information.

- 5.1. **Compliance:** The solution must comply with data protection standards and all relevant government regulations.
- 5.2. **Framework Adherence:** The Consultant must adhere to the Disclosure and Accountability Framework as stipulated by the Procuring Agency.
- 5.3. **Security Requirements:** The system must implement robust security features and must pass security audits as outlined in the Non-Functional Requirements.

## 6. Timeline, Milestones, and Payment

This is a time-bound project and all the development work and testing have to be completed **sixty [60]** business days from signing the contract. Consultants should propose a realistic timeline that aligns with milestones, clearly indicating their proposed durations and any dependencies. Any deviations from this target timeline must be justified in their proposal. Payment will be disbursed in milestone-based tranches aligned with deliverables and performance verification. Indicative milestones include:

| Milestone | Description                                                          | Estimated Duration | Payment %   |
|-----------|----------------------------------------------------------------------|--------------------|-------------|
| M1        | Detailed Requirements Analysis System Design & Architecture Approval | D + W2             | 15%         |
| M2        | Development & Unit Testing, Quality Assurance & UAT                  | D + W5             | 40%         |
| M3        | Deployment, Go-Live, and Handover                                    | D + W6             | 35%         |
| M4        | Post-Implementation Support                                          | D + W8             | 10%         |
|           |                                                                      |                    | <b>100%</b> |

Ten percent (10%) of each milestone payment shall be withheld as a Performance Guarantee and released only after the warranty period, contingent on the consultant providing timely maintenance, vulnerability fixes, and updates; non-compliance will forfeit the retained amount.

## 7. Roles and Responsibilities

### 7.1. Procuring Agency:

- 7.1.1. Providing timely access to necessary documentation, systems, and personnel.
- 7.1.2. Reviewing and approving deliverables within agreed-upon timelines.
- 7.1.3. Facilitating user acceptance testing (UAT) and providing constructive feedback.
- 7.1.4. Provide consultants with the official GitLab account and hosting for the project.
- 7.1.5. Making timely payments as per the agreed contract.

### 7.2. Consultant:

- 7.2.1. Adhering to the project scope, objectives, and timelines outlined in this ToR.
- 7.2.2. Delivering in accordance with the functional and non-functional requirements.
- 7.2.3. Ensuring the quality, security, and performance of the developed solution.
- 7.2.4. Providing comprehensive documentation and training.
- 7.2.5. Maintaining regular communication and reporting project progress.
- 7.2.6. Adhering to industry best practices and coding standards.
- 7.2.7. Adhere to the Disclosure and Accountability Framework.
- 7.2.8. Maintain a Risk Register throughout the project lifecycle.
- 7.2.9. Accommodate up to 20% of change requests to the total cost during the development phase without additional cost.

## 8. Evaluation Criteria

Proposals received will be evaluated based on the following criteria. Consultants are encouraged to address each of these criteria comprehensively in their proposals. At least 70% of the total team must be Bhutanese nationals.

Technical score 70% and financial score 20%

| # | Category                                                                                                 | Criteria                                                          | Description                                                                                          | Weight |
|---|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|
| 1 | Consultant's Organisation and Experience                                                                 | Specific Experience of the Consultant Relevant to the Assignment. | E.g.: similar nature, scale, context.                                                                | 10     |
|   |                                                                                                          | General Experience of the Consultant.                             | E.g.: Professional Tenure, Number of Projects in Hand( See #10: Additional Required Form Submission) | 5      |
| 2 | Comments or Suggestions on the ToR and on counterpart staff and facilities to be provided by the client. | Content Presentation                                              | Presentation / Demo Evaluation                                                                       | 10     |
| 3 | Description of Approach, Methodology and Work Plan in Responding to the ToR                              | Technical Approach and Methodology                                | E.g.: Approach, Methodology, diagrams, etc                                                           | 20     |
|   |                                                                                                          | Work Plan                                                         | E.g. Work plan, timeline, deliveries,etc                                                             | 05     |
|   |                                                                                                          | Organisation and Staffing                                         | E.g.: Staffs, Roles, Hierarchy, etc                                                                  | 15     |
| 4 | Work Schedule and planning for Deliverables                                                              | Others                                                            | Fill up the provided templates in the eGP.                                                           | 3      |
| 5 | Curriculum Vitae                                                                                         | Educational Qualification                                         | E.g.: Masters, Undergrade, Diploma, etc                                                              | 06     |
|   |                                                                                                          | Relevant Working Experience and its adequacy for the Assignment   | E.g.: considering schedule, experience, etc                                                          | 17     |
|   |                                                                                                          | Suitability considering skills and Trainings                      | E.g.: Relevant Trainings, Certifications, etc to Role (e.g: PMP)                                     | 7      |
| 6 | Team Composition, Assignment and key expert's inputs                                                     | Others                                                            | Fill up the provided templates in the eGP.                                                           | 2      |

Total 100

## 9. Additional Mendatory Form Submission

### 9.1. Number of Projects in Hand (Workload)

| Project                                              | Type                                         | Start      | End        | System Description                                                                                                                                                      | Tech Stack                             | Client Details                                       | Any of the proposed staff involved in this project?                                        |
|------------------------------------------------------|----------------------------------------------|------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Mobile App/<br>Web-Based/<br>Database/<br>System App | Warranty/<br>Support/<br>AMC/<br>Development | dd/mm/yyyy | dd/mm/yyyy | Objective/Modules/Industry                                                                                                                                              | Language/DB/O<br>S/Server              | Brief Description/<br>Contact                        | ID, Name, Role in project,<br>Contact                                                      |
| E.g.,<br>Web-base                                    | Development                                  | 01/12/2024 | 01/12/2025 | For Hospital Administration to improve issuance of medical documents. The system has 12 modules. Includes Digital Signature, Document Storage, and Approval [ More ...] | Java/Net,<br>PgSQL/Frame<br>works, etc | www.example-c.co<br>m<br>ex@test.com<br>975-23231244 | ID:1111111111<br>Name: Mr. Dorji Dawa,<br>Role: Front-end Developer<br>Contact: 975-123123 |
|                                                      |                                              |            |            |                                                                                                                                                                         |                                        |                                                      |                                                                                            |

\*\*\* END of ToR \*\*\*